

Titanic Survival Analysis Report

1. Introduction

This report presents an exploratory data analysis of the Titanic dataset. The objective of this analysis is to understand which factors influenced passenger survival. By examining variables such as passenger class, fare, port of embarkation, and family size, we aim to identify patterns that affected survival outcomes.

2. Dataset Description

The Titanic dataset contains information about passengers aboard the Titanic ship. The dataset includes several variables such as Passenger Class (Pclass), Age, Sex, Fare, number of siblings/spouses aboard (SibSp), number of parents/children aboard (Parch), and the port of embarkation (Embarked). The target variable is Survival, which indicates whether a passenger survived the disaster.

3. Methodology

In this project, Exploratory Data Analysis (EDA) was conducted using Python. The Pandas library was used for data manipulation and grouping, while Matplotlib was used for creating visualizations. Survival rates were calculated by grouping data according to different variables such as fare groups, embarkation ports, and family-related features (SibSp and Parch). Bar charts were used to visualize these relationships.

4. What I found?!

The dataset contains 891 records and 11 columns describing Titanic passengers. Most columns such as Survived, Pclass, Name, Sex, SibSp, Parch, Ticket, and Fare have complete data. However, some columns contain missing values. The Age column has 177 missing values, the Cabin column has 687 missing values, and the Embarked column has 2 missing values. These missing values need to be handled during the data preprocessing stage.

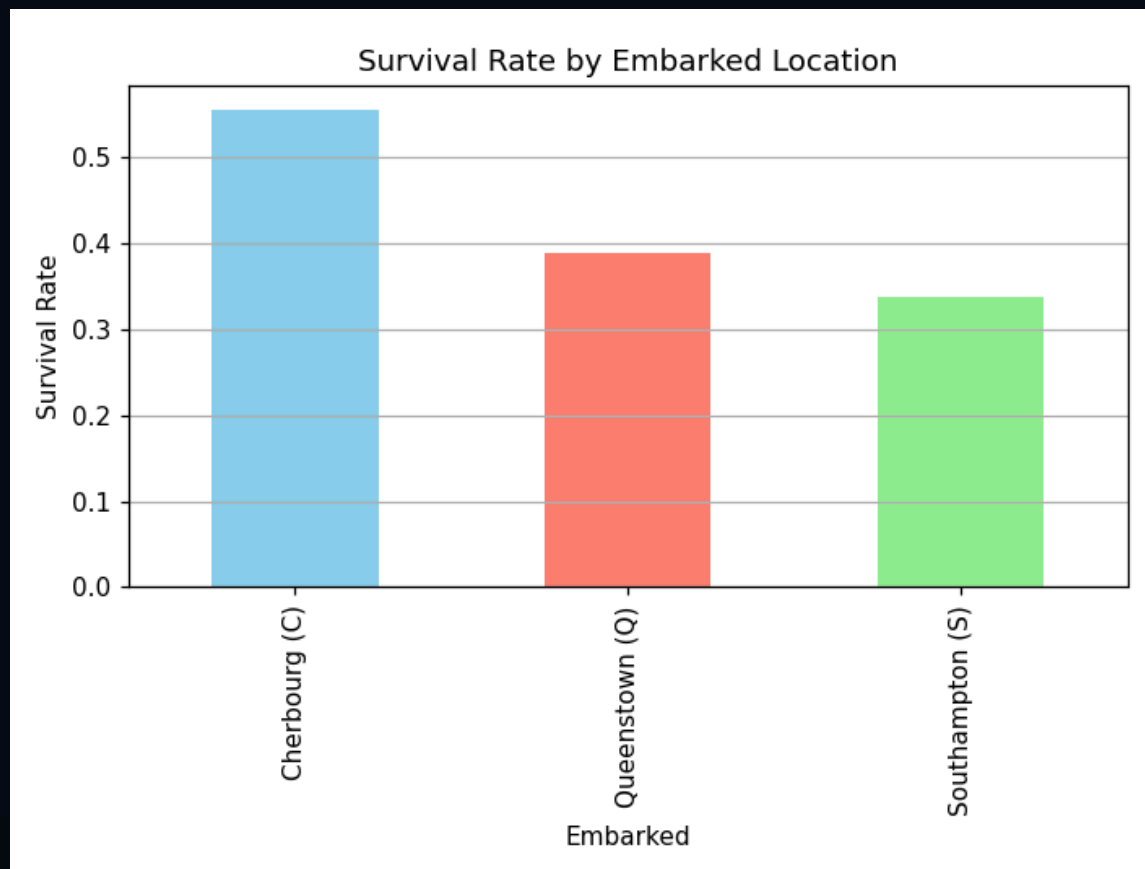
5. Handling Missing Values

To handle the missing data, different strategies were applied. The missing values in the Age column were replaced using the median age to avoid the effect of extreme values. The Cabin column was removed from the dataset because it contained a large number of missing values (more than half of the entries), which could reduce the accuracy of the analysis. For the Embarked column, the two missing values were filled based on the embarkation port of similar nearby records, assuming they likely embarked from the same port.

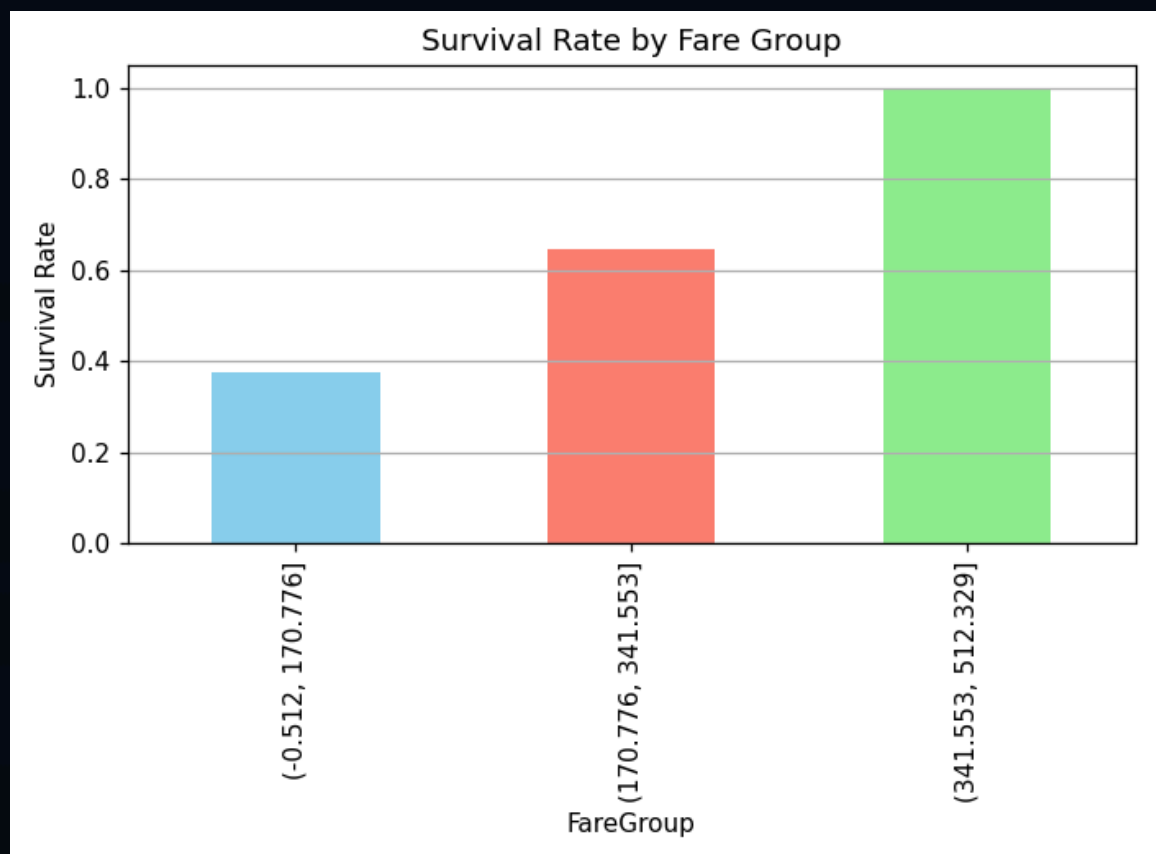
4. Results and Findings

The analysis revealed several insights:

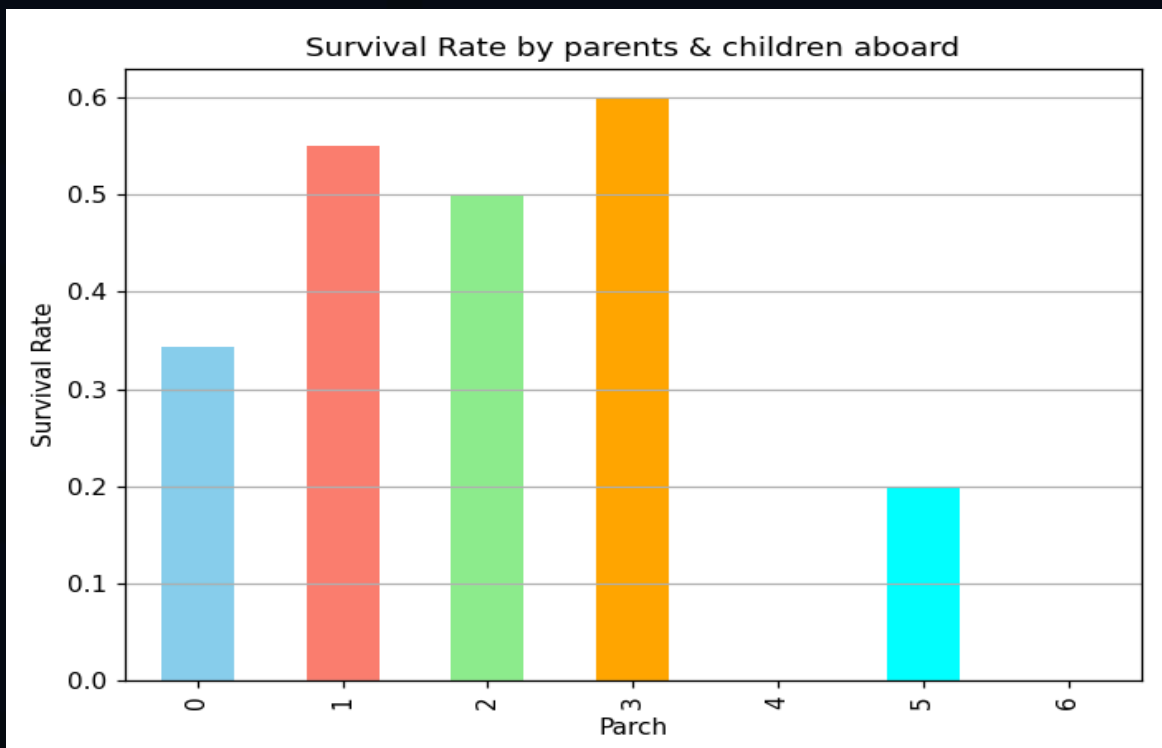
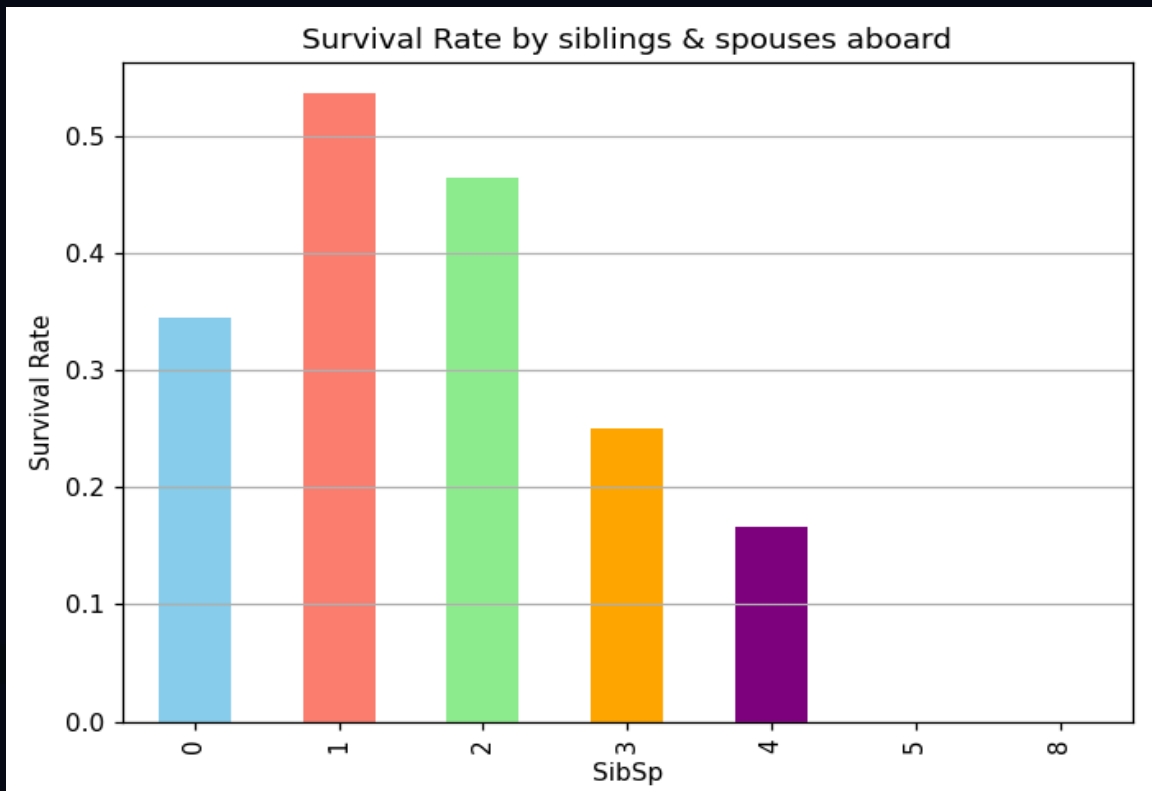
1- Passengers who embarked from Cherbourg had a higher survival rate compared to those from Southampton and Queenstown:



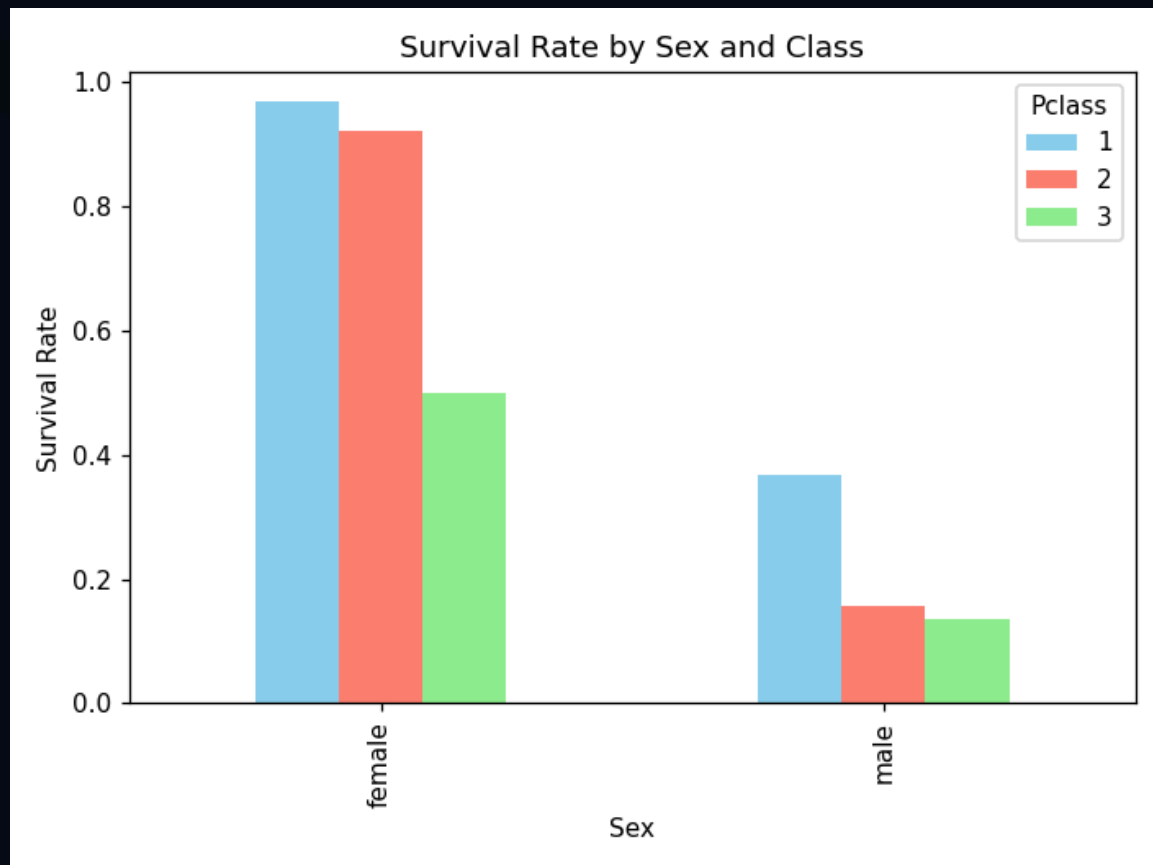
2- Survival rates varied across fare groups, suggesting that passengers who paid higher fares had better chances of survival:



3- Passengers traveling with fewer family members generally showed slightly higher survival rates.



4-Female passengers had much higher survival rates than males in all passenger classes. First-class females had the highest survival rate, while survival rates decreased in lower classes, especially for male passengers.



5. Conclusion

In conclusion, the Titanic dataset provides valuable insights into the factors that influenced survival during the disaster. Variables such as passenger class, fare, and port of embarkation appear to have played significant roles. Data visualization helped reveal these patterns clearly. Future work could include building machine learning models to predict survival probabilities based on these features.